

Total No. of Questions : 10]

SEAT No. :

P1355

[Total No. of Pages : 2

[4858] - 1103

T.E. (Information Technology) (Semester - I)

End Semester Examination

COMPUTER NETWORK TECHNOLOGY

(2012 Pattern)

Time : 3 Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Neat diagrams must be drawn wherever necessary.*
- 2) Figures to the right side indicate full marks.*
- 3) Assume suitable data if necessary.*

Q1) a) How token bucket algorithm works in case of smooth and bursty traffic. [6]

b) What is classless addressing? Explain. [4]

OR

Q2) a) Differentiate between IPv4 and IPv6. [4]

b) A company is assigned site address 140.55.0.0. The company needs 1000 IP address. Find the no. of subnet and range of IP address of each subnet. [6]

Q3) a) State the ICMP message format and list ICMP message types. [6]

b) Explain Mobile IP. [4]

OR

Q4) a) Explain working of DHCP with transition diagram. [6]

b) How to overcome the problems in RIP? [4]

Q5) a) Write characteristics of radio propagation and propagation impairments. [8]

b) Explain Bluetooth architecture and its protocol stack. [10]

OR

P.T.O.

- Q6)** a) Write short note on : (any two) [10]
i) LEACH
ii) AODV
iii) DSR
b) Explain IEEE 802.11 FHSS and DSSS. [8]

- Q7)** a) Explain the basic architecture of WLAN and discuss various components in it. [8]
b) Discuss naming and addressing in wireless sensor network. [8]

OR

- Q8)** a) State different MAC protocols in sensor network? Explain S-MAC in detail. [8]
b) What are different design issues and challenges of wireless sensor network? [8]

- Q9)** a) Explain in detail CSMA/CA. [8]
b) Differentiate between content based and geographic routing. [8]

OR

- Q10)** a) Explain SPIN routing protocol for WSN. [8]
b) What are the advantages and disadvantages of BYOD? [8]

